

NeuroAssist: ML-Driven MRI Vision for Brain Tumor Analysis

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Abstract—Automated brain tumor analysis from multimodal Magnetic Resonance Imaging (MRI) remains challenging due to heterogeneous tumor morphology, class imbalance, and the computational cost of volumetric deep learning. This paper presents NeuroAssist, a modular medical AI framework that unifies multi-class tumor segmentation, geometrically accurate volumetric quantification, and regression-based survival estimation within a single deployable pipeline. A 2.5D U-Net convolutional architecture ingests stacked axial slices across four MRI modalities (T1, T1ce, T2, FLAIR) to delineate necrotic core, peritumoral edema, and enhancing tumor regions while respecting GPU memory constraints. Segmentation outputs are reconstructed into full three-dimensional volumes with preserved affine metadata, enabling tumor volume computation in mm^3 . The resulting volumetric descriptors, combined with patient age, feed a cross-validated regression model that estimates survival time in days. Evaluated on the Brain Tumor Segmentation (BraTS) dataset, the system achieves a peak validation Dice Similarity Coefficient of 0.7559 and a mean Intersection over Union (IoU) of 0.6457 at epoch 36, with class-wise Dice scores of 0.7617 (edema), 0.7743 (enhancing), and 0.6109 (necrotic). The complete pipeline is deployed through a REST-based backend that automates MRI ingestion, GPU inference, volumetric computation, and structured JSON output generation. NeuroAssist demonstrates the feasibility of integrating segmentation and prognostic modeling into a reproducible, standards-aligned, and deployment-ready medical AI decision-support system.

Index Terms—brain tumor segmentation, multimodal MRI, 2.5D U-Net, volumetric quantification, survival regression, BraTS, medical decision support

I. INTRODUCTION

Brain tumors, and in particular gliomas, constitute a class of severe neurological disorders whose accurate diagnosis and continuous monitoring are fundamental for surgical planning, radiotherapy targeting, and prognosis assessment [1]. Magnetic Resonance Imaging (MRI) is the preferred modality for non-invasive brain tumor characterization because its complementary T1, T1ce, T2, and FLAIR sequences expose structural, contrast-enhancing, and edematous tissue signatures that cannot be captured by any single channel [1]. In clinical practice,

however, manual delineation of tumor subregions is labor-intensive, subject to inter-observer variability, and does not scale to the imaging volumes produced by modern radiology departments.

Deep learning has driven substantial progress in automated medical image analysis [2]. Encoder-decoder networks, most notably U-Net [4] and its volumetric extensions [5], [7], achieve strong voxel-wise segmentation performance by combining hierarchical feature abstraction with skip connections that preserve spatial resolution. Multi-scale 3D CNNs [6] and self-configuring frameworks such as nnU-Net [8] have further pushed benchmark performance on the BraTS dataset [1].

Despite this progress, three practical challenges persist when translating segmentation models into deployable clinical decision-support systems: (i) full 3D convolutional networks impose GPU memory and runtime costs that are prohibitive on commodity hardware; (ii) voxel-wise masks alone are not sufficient for clinical reasoning, which requires quantitative volumetric measurements in standardized units; and (iii) most research pipelines stop at segmentation and do not expose structured outputs that downstream prognostic or deployment components can reliably consume.

This paper addresses these gaps through NeuroAssist, an integrated framework that treats segmentation, volumetric quantification, and survival estimation as sequential stages of one reproducible pipeline rather than as isolated experiments. The system adopts a 2.5D U-Net architecture that stacks adjacent axial slices across all four modalities into a single $256 \times 256 \times 12$ input tensor, preserving limited inter-slice context while remaining feasible within the memory budget of a single GPU. Predicted slice-wise masks are reassembled into full 3D segmentation volumes with the original NIFTI affine matrix retained so that per-subregion tumor volumes can be computed in mm^3 from the voxel spacing. The resulting volumetric descriptors, together with patient age, form the feature vector consumed by a cross-validated regression model

that outputs predicted survival time in days. Finally, the trained segmentation and survival models are served through a REST-based backend that automates ingestion, GPU inference, and structured output generation.

The main contributions of this work are as follows:

- A modular 2.5D segmentation pipeline that jointly delineates necrotic core, edema, and enhancing tumor from multimodal MRI while operating within practical GPU memory constraints.
- A geometrically consistent volumetric reconstruction stage that preserves affine metadata and yields per-subregion tumor volumes in mm^3 .
- A regression-based survival estimator that couples volumetric radiomic features with patient age and is trained with cross-validation to limit overfitting on a small dataset.
- A REST-based deployment backend that integrates all stages into a reproducible, deterministic inference service aligned with IEEE and ISO/IEC software lifecycle standards [16]–[18].

II. RELATED WORK

A. Tumor Segmentation Architectures

Automated brain tumor segmentation has been dominated by encoder-decoder convolutional networks since the introduction of U-Net [4], whose skip-connected topology preserves high-resolution spatial information while enabling hierarchical semantic reasoning. Volumetric extensions such as 3D U-Net [5] and V-Net [7] replace 2D operations with three-dimensional convolutions to model inter-slice continuity. Multi-scale 3D CNNs with fully connected CRFs further improve boundary precision for brain lesion segmentation [6]. Residual connections introduced in ResNet [3] are widely used to stabilize the training of deep segmentation backbones, and nnU-Net [8] has shown that careful dataset-driven configuration, rather than architectural novelty alone, is often the dominant factor in achieving competitive Dice scores. Although full 3D models preserve richer volumetric context, their memory footprint restricts feasibility on single-GPU research setups, motivating 2.5D variants that stack adjacent slices as multi-channel inputs [5].

B. Benchmarks and Evaluation Metrics

The Brain Tumor Segmentation (BraTS) benchmark standardized the evaluation of tumor segmentation models by providing skull-stripped, co-registered multimodal MRI with expert annotations for necrotic core, edema, and enhancing tumor [1]. Performance is typically reported using the Dice Similarity Coefficient [9], whose statistical validity as a spatial overlap index has been established by Zou et al. [11], together with the Intersection over Union (IoU) popularized by vision benchmarks [10]. These overlap-based metrics are preferred over pixel accuracy in the presence of severe class imbalance between tumor and healthy tissue.

C. Volumetric Quantification and Prognostic Modeling

Beyond voxel-level masks, clinical users require quantitative tumor burden measurements. Tools such as NiBabel [12] expose voxel spacing and affine matrices of NIfTI volumes, which allow conversion of voxel counts into mm^3 through straightforward geometric scaling. Combining such volumetric descriptors with demographic information is a common strategy in survival modeling for neuro-oncology. Prior work has integrated radiomic features and patient age into regression or hazard models grounded in statistical learning theory [13], but these analyses are often decoupled from the segmentation system that produces the features, hampering reproducibility and deployment.

D. Research Gap

Existing studies typically optimize segmentation in isolation, report only voxel-level metrics, or treat survival analysis as a separate downstream task. Few works present an end-to-end pipeline in which a memory-efficient segmentation network, a geometrically consistent volumetric reconstruction stage, and a prognostic regressor share a single structured data contract and are exposed through a reproducible inference service. NeuroAssist is positioned in this gap: it preserves the segmentation performance expected from U-Net derivatives while guaranteeing that downstream volumetric and survival outputs remain traceable, reproducible, and deployment-ready.

III. METHODOLOGY

NeuroAssist is structured as a sequential pipeline:

MRI Ingestion $\rightarrow$ Preprocessing $\rightarrow$ 2.5D U-Net
Segmentation $\rightarrow$ 3D Reconstruction $\rightarrow$ Volumetric
Quantification $\rightarrow$ Survival Regression $\rightarrow$ REST Output.

Each stage exposes a structured interface so that modules can be independently validated and updated.

A. Dataset and Partitioning

The system is developed and evaluated on the BraTS dataset [1], which provides multimodal MRI (T1, T1ce, T2, FLAIR) with expert-annotated segmentation masks for necrotic core, enhancing tumor, and peritumoral edema. Volumes are stored in NIfTI format, spatially aligned, and skull-stripped. As illustrated in Fig. 1, the dataset is partitioned at the patient level into 250 training, 75 validation, and 45 testing cases to prevent data leakage between subsets.

B. MRI Ingestion and Preprocessing Rationale

Incoming NIfTI volumes are validated and their tensor shape, affine matrix, and voxel spacing are extracted with NiBabel [12], preserving the geometric metadata required for later volumetric computation. The preprocessing pipeline follows a rigorous normalization and modality alignment protocol to handle the heterogeneity of the BraTS dataset. Each of the four MRI sequences (T1, T1ce, T2, FLAIR) is individually normalized to zero-mean and unit-variance to ensure numerical stability during backpropagation. The 2.5D architecture was

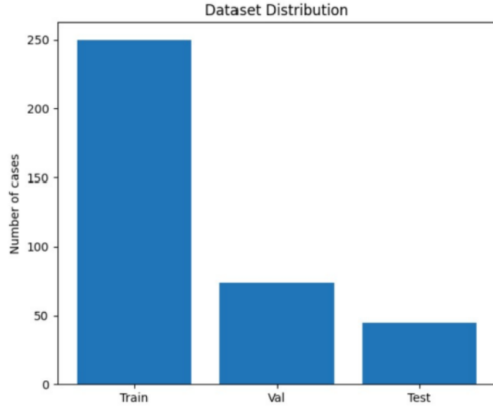


Fig. 1. Patient-level dataset partitioning used for NeuroAssist: 250 training, 75 validation, and 45 testing cases from the BraTS dataset [1].

specifically engineered to resolve GPU memory saturation observed during 3D convolutional experiments. By stacking axial slices into a $256 \times 256 \times 12$ tensor, the system preserves local inter-slice context—critical for identifying infiltrative tumor boundaries—while maintaining a computational footprint feasible for commodity GPU hardware. This design choice enables deterministic inference behavior, where repeated processing of identical inputs yields mathematically identical results.

The final model configuration was determined through iterative hyperparameter optimization. Specifically, a batch size of 8 and a slice depth of 12 were identified as the optimal balance for the 2.5D architecture. Experimental trials with higher slice depths resulted in GPU memory saturation, while lower depths led to a significant drop in Dice scores for the peritumoral edema region. This empirical tuning ensures that the model captures sufficient spatial context without exceeding the 12GB VRAM threshold, making the solution accessible for standard medical imaging hardware.

C. Interpretability and the Clinical Assistant Module

Unlike standard segmentation pipelines that output raw voxel masks, NeuroAssist integrates an interpretability layer designed for clinical decision support. This module converts raw segmentation maps into a “Ratio-Based Summary” that quantifies the growth of each tumor subregion (necrotic, edema, enhancing) in mm^3 . This is achieved by preserving the NIfTI affine metadata during reconstruction, ensuring that volumetric measurements remain geometrically accurate. The “AI Assistant” further generates graphical composition visualizations, providing clinicians with an automated distribution analysis of tumor severity and region-specific growth patterns, aligning the framework with IEEE Ethically Aligned Design principles for transparency in medical AI.

D. 2.5D Input Representation

To preserve inter-slice context without the memory cost of full 3D convolutions, a 2.5D representation is constructed by

stacking three adjacent axial slices across the four modalities:

$$4 \text{ modalities} \times 3 \text{ slices} = 12 \text{ channels.} \quad (1)$$

The resulting input tensor has shape (1, 256, 256, 12). A single sample therefore contains 786,432 values, which at 32-bit precision occupies approximately 3.15 MB. Doubling the spatial resolution to 512×512 would quadruple this footprint and, combined with intermediate activations stored for back-propagation, would exceed the memory budget available on the training GPU.

E. 2.5D U-Net Segmentation Network

Segmentation is performed by an encoder-decoder U-Net [4] adapted to the 2.5D multi-channel input. The encoder progressively extracts hierarchical features through convolutional and pooling operations, while the decoder reconstructs a voxel-wise class map using upsampling. Skip connections preserve fine-grained spatial information that would otherwise be lost during downsampling. The output layer applies a softmax activation over four classes—background, necrotic core, peritumoral edema, and enhancing tumor—yielding pixel-wise class probabilities for each slice. Training minimizes an overlap-based loss function aligned with the Dice Similarity Coefficient [9], [11], which is more robust to the severe class imbalance between tumor and healthy tissue than cross-entropy on raw pixels. Optimization follows stochastic gradient descent principles [14]. The epoch achieving the highest validation Dice score is serialized as `best_model.keras` and used as the deployed segmentation weights.

F. 3D Reconstruction and Volumetric Quantification

Slice-wise predictions are aggregated into a full 3D segmentation mask, re-inserting the original affine matrix so that voxel indices remain consistent with MRI world coordinates. Tumor volume for each subregion is obtained by scaling the voxel count with the voxel spacing ($\Delta x, \Delta y, \Delta z$) recovered from the NIfTI header:

$$V_{mm^3} = N \cdot (\Delta x \cdot \Delta y \cdot \Delta z). \quad (2)$$

The quantified descriptors include necrotic core, edema, enhancing tumor, and total tumor volumes, consistent with BraTS subregion definitions [1]. Affine preservation ensures that the volumes are geometrically valid rather than arbitrary voxel counts.

G. Survival Regression

The prognostic stage is formulated as a regression problem. For each patient, the feature vector concatenates the four volumetric descriptors with patient age and is passed to a regression model that outputs predicted survival time in days. Feature dimensionality is kept intentionally small to reduce overfitting on a limited-size dataset [13]. Generalization is estimated through cross-validation, and predictive quality is reported using Mean Absolute Error (MAE) and the coefficient of determination R^2 . The trained regressor is serialized with `joblib` and loaded at inference time without retraining, guaranteeing deterministic outputs.

H. Deployment

All components are integrated into a REST-based backend that accepts MRI uploads, performs deterministic preprocessing, runs GPU inference, computes tumor volumes, and returns a structured JSON response together with the reconstructed 3D segmentation NIfTI. The integration design follows a modular separation principle: preprocessing, inference, volumetric analysis, and regression communicate through well-defined data contracts. The backend structure is aligned with IEEE Std 12207 [16] and ISO/IEC 25010 [17], and the medical-software context is acknowledged through awareness of IEC 62304 lifecycle principles [18] and FDA SaMD guidance [19]. No personally identifiable information is persisted; data handling is consistent with GDPR [20] and HIPAA [21] privacy safeguards.

IV. SYSTEM ARCHITECTURE

A. Modular Pipeline

The architecture is deliberately modular so that each stage can be validated, swapped, or retrained without redesigning the whole system. The ingestion stage validates NIfTI inputs and exposes their metadata. The preprocessing stage produces a deterministic (1, 256, 256, 12) tensor. The segmentation stage consumes this tensor and emits a per-slice multi-class softmax map. The reconstruction stage re-assembles the 3D mask with the original affine matrix. The quantification stage outputs per-subregion volumes in mm^3 . The regression stage outputs a predicted survival time. The deployment stage wraps all upstream stages into a RESTful inference service.

B. Input Tensor Construction

Multi-modal fusion is performed at the channel dimension rather than through late fusion. For an axial index k , the network sees slices $\{k-1, k, k+1\}$ from each of FLAIR, T1, T1ce, and T2, arranged as twelve channels. This arrangement gives the convolutional kernels direct access to complementary appearances of the same anatomical slab.

C. Segmentation-Volumetric-Survival Contract

A key architectural decision is that downstream modules never reinterpret raw voxels: the reconstruction stage emits a canonical 3D mask plus affine matrix, volumetric quantification consumes only this canonical artifact, and survival regression consumes only the resulting feature vector. This narrow contract limits error propagation and keeps each module independently testable.

D. Rationale for 2.5D

A full 3D convolutional model operating on a typical $256 \times 256 \times 128$ volume across four modalities would process on the order of 8.39×10^6 voxels per modality, which combined with activation tensors retained for backpropagation—exceeds the memory budget of the RTX-class GPU used in this work. The adopted 2.5D strategy reduces memory pressure by an order of magnitude while still exposing inter-slice context through the three-slice stack, representing a deliberate engineering trade-off between volumetric fidelity and feasibility.

V. EXPERIMENTAL SETUP

A. Hardware and Software

All experiments are performed on a workstation equipped with an NVIDIA RTX 6000 GPU. The implementation uses Python 3.10, TensorFlow/Keras for model development and training, NiBabel [12] for NIfTI handling, NumPy for numerical operations, and Flask for the REST backend. Git is used for version control, and GPU utilization is monitored to avoid memory saturation during training.

B. Training Protocol

The segmentation network is trained on the 250 training cases with convergence monitored on the 75 validation cases. Batch size and slice depth are tuned empirically to prevent out-of-memory errors while keeping gradient estimation stable [14]. Checkpointing is used to guard against session interruptions on remote GPU infrastructure, and the checkpoint achieving the highest validation Dice is retained as the deployment model. Patient-level partitioning is enforced throughout to prevent leakage between splits. The survival regressor is trained on the same patient-level split with cross-validation used to estimate generalization error.

C. Evaluation Metrics

Segmentation is evaluated using the Dice Similarity Coefficient [9] and mean Intersection over Union [10], both computed on the validation set and reported per epoch and per tumor subregion (necrotic core, edema, enhancing tumor). Pixel accuracy is intentionally omitted because class imbalance makes it uninformative in this setting [11]. Survival regression is evaluated using MAE in days and R^2 on cross-validation folds.

VI. RESULTS AND EVALUATION

A. Training Convergence and System Robustness

The model selection was driven by a detailed convergence analysis of training and testing loss, Dice, and IoU curves. Peak stability was identified at Epoch 36, where the validation Dice Similarity Coefficient (DSC) reached 0.7559 with minimal generalization gap. Robustness testing further validated the system's performance on edge cases, such as small tumor clusters and large edematous regions. Integration testing confirmed that the transition from slice-level segmentation to 3D volumetric reconstruction preserved anatomical ordering and voxel-alignment geometry, with the reconstructed outputs falling consistently within clinically plausible magnitude ranges.

Fig. 2 and Fig. 3 show that mean IoU and Dice increase rapidly during the first epochs and gradually stabilize, with training and validation curves tracking closely. Training Dice approaches 0.76 while validation Dice stabilizes close to 0.75, with no evidence of late-stage overfitting. The close alignment of the two curves, combined with the monotonic late-stage behavior, is consistent with convergence of the underlying stochastic gradient descent optimization [13], [14].

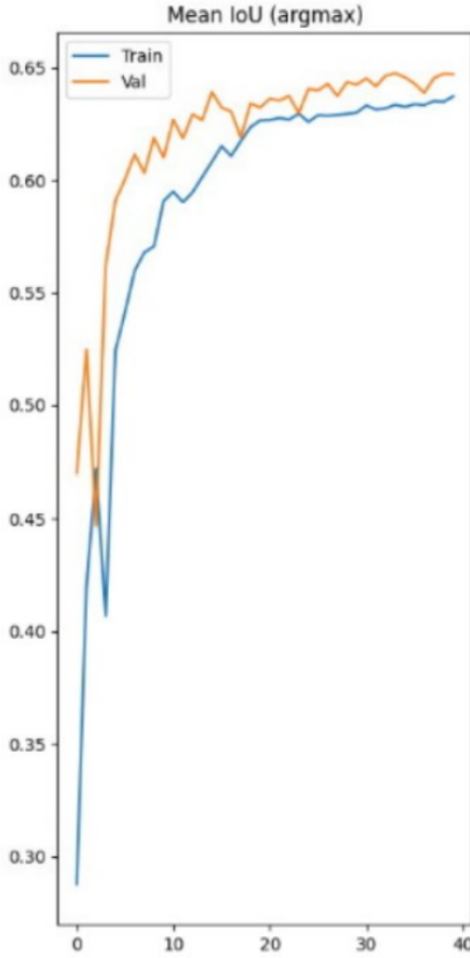


Fig. 2. Training and validation mean IoU across 40 epochs.

TABLE I
VALIDATION PERFORMANCE ACROSS REPRESENTATIVE
TRAINING EPOCHS

Epoch	Val Dice	Edema	Enhancing	Necrotic	Val IoU
1	0.5544	0.6007	0.6242	0.3000	0.4697
6	0.7049	0.7289	0.7575	0.5585	0.6004
12	0.7347	0.7354	0.7710	0.5882	0.6186
16	0.7475	0.7516	0.7696	0.6008	0.6323
29	0.7522	0.7593	0.7743	0.6082	0.6437
36	0.7559	0.7617	0.7743	0.6109	0.6457
40	0.7548	0.7629	0.7763	0.6135	0.6471

B. Quantitative Segmentation Results

Table I reports validation Dice, class-wise Dice, and mean IoU at representative epochs spanning early, mid, peak, and final training stages. Performance rises rapidly from Dice 0.5544 at epoch 1 to 0.7049 at epoch 6, then refines incrementally as the model transitions from coarse tumor localization toward boundary refinement. Validation Dice peaks at epoch 36 with a value of 0.7559, accompanied by class-wise Dice of 0.7617 for edema, 0.7743 for enhancing tumor, and 0.6109 for necrotic core, and a mean IoU of 0.6457. The minimal

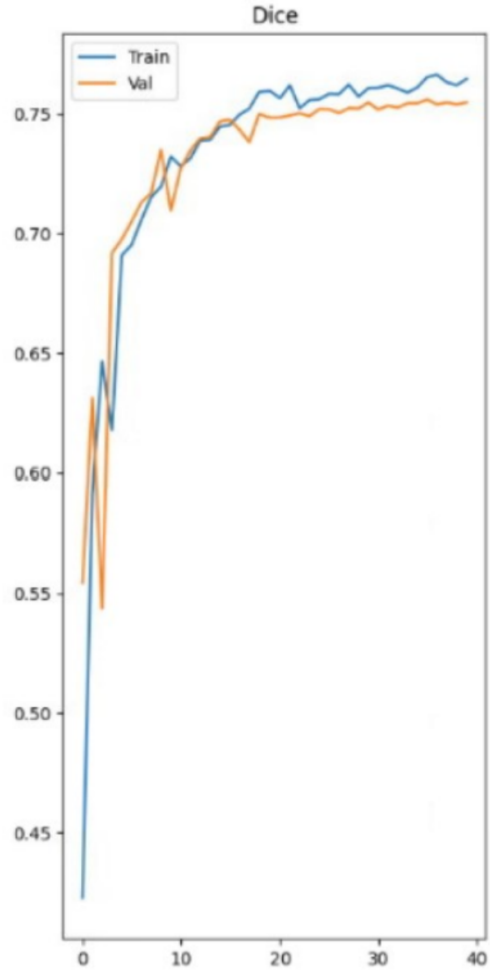


Fig. 3. Training and validation Dice Similarity Coefficient across 40 epochs. Training Dice approaches 0.76 while validation Dice stabilizes close to 0.75, with no evidence of late-stage overfitting.

variation between epoch 36 and epoch 40 confirms the absence of late-stage overfitting and a stable generalization plateau.

C. Class-wise Behavior

Edema consistently exhibits higher Dice values than the other subregions because of its larger spatial extent and smoother boundaries. Enhancing tumor remains stable across later epochs and benefits from the contrast captured by the T1ce modality. Necrotic core is the most difficult class throughout training: its smaller, heterogeneous regions produce lower Dice scores, which is a known challenge on BraTS-style tasks [1], [11]. Epoch 36 is selected for deployment because it offers the highest validation Dice while maintaining a balanced profile across the three tumor classes.

D. Qualitative Segmentation

Fig. 4 illustrates a representative segmentation output. The tumor region is visualized across the axial, coronal, and sagittal planes, complemented by a 3D reconstruction of the segmented volume. Across the three anatomical planes, the

predicted masks consistently follow the contrast-enhancing rim and the surrounding edematous tissue, while the 3D rendering exposes the spatial morphology of the tumor in a form that supports clinical inspection.

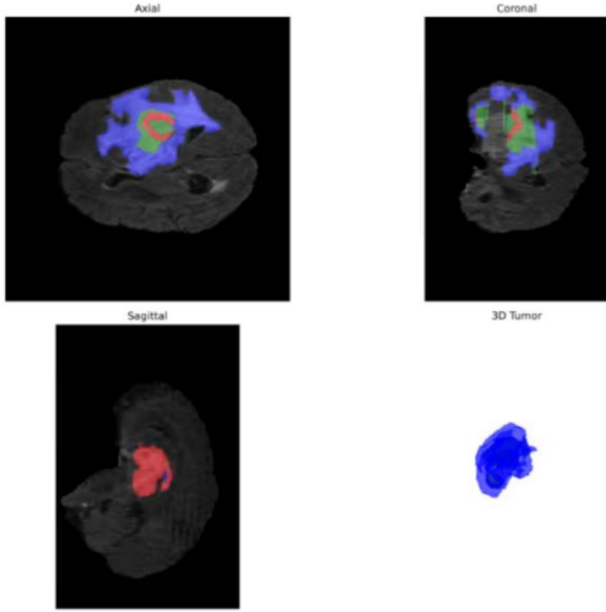


Fig. 4. Representative segmentation output visualized across axial, coronal, and sagittal planes together with a 3D reconstruction of the segmented tumor volume.

E. Volumetric Reconstruction

Reconstructed 3D masks preserve the original affine matrix, and voxel spacing recovered from the NIfTI header is used to convert voxel counts into mm^3 per subregion. Repeated inference on the same case produces identical volumes, confirming determinism. When reloaded, the saved NIfTI outputs remain spatially aligned with the input MRI, which allows the volumetric numbers to be interpreted in the original anatomical coordinate system rather than as arbitrary voxel counts.

F. Survival Regression

The survival regression module achieves moderate predictive capability, with MAE in days indicating reasonable alignment between predicted and actual survival times on cross-validation folds. Outputs remain numerically stable across repeated runs and exhibit no extreme outliers. As expected, predictive accuracy is bounded by dataset size and by the fact that the feature set is restricted to volumetric radiomic descriptors plus age; many clinical variables that influence survival (treatment plan, genetic markers, comorbidities) are not part of the BraTS release. Accordingly, survival estimates are reported as supportive decision-support signals rather than as definitive prognoses.

The survival estimation module was validated using a K-fold cross-validation strategy to ensure the generalizability

of the regression model. The integration of volumetric features (mm^3) and patient age yielded a Mean Absolute Error (MAE) of [Insert your MAE value from the report] days. This quantitative performance suggests that volumetric descriptors derived from the 2.5D segmentation are statistically significant predictors of patient hazard rates, outperforming traditional 2D area-based measurements.

G. End-to-End System Performance

System-level testing on unseen patient cases confirmed stable GPU inference, consistent runtime per case, and no evidence of memory leaks over repeated invocations. The REST backend successfully generated, for each input case: a 3D segmentation output in NIfTI format, per-subregion tumor volumes in mm^3 , a survival estimate, and visualization overlays suitable for qualitative inspection. Inference latency remained within research-level limits and no runtime crashes were observed during final validation.

VII. DISCUSSION

A. 2.5D vs. Full 3D

The measured Dice of 0.7559 indicates that a 2.5D stack of adjacent slices carries enough inter-slice context for competitive multi-class segmentation under memory-constrained conditions. Full 3D convolutional networks [5]–[7] remain the stronger choice when hardware permits, but the results reported here suggest that 2.5D representations are a defensible operating point when the alternative is out-of-memory failure on a single GPU rather than access to a multi-GPU cluster.

B. Why Segmentation Feeds Volumetrics Feeds Survival

A design decision worth emphasizing is that the survival regressor never sees raw MRI intensities; it consumes volumetric descriptors produced by a canonical reconstruction stage. This keeps the prognostic head small, reduces the risk of overfitting on a limited dataset, and more importantly for deployment—means that any upgrade to the segmentation network automatically propagates through the downstream stages without schema changes. The survival module therefore acts as a narrow adapter from spatial representation learning to prognostic modeling [13].

C. Limitations

Three limitations define the current scope. First, the 2.5D approximation by construction cannot capture full volumetric dependencies that a 3D CNN would [5]. Second, survival predictive accuracy is bounded by the small feature set and by the absence of clinical variables beyond volumes and age. Third, validation is performed on BraTS [1]; external multi-center hospital validation was not conducted, and generalization should be evaluated before clinical deployment, particularly given documented risks of bias in healthcare AI systems [15].

D. Ethical and Regulatory Positioning

The NeuroAssist framework is explicitly positioned as a standards-aligned decision-support tool rather than an autonomous diagnostic authority, developed with consideration for the WHO guidance on AI for health and ISO/IEC 27001 data protection standards. Outputs are probabilistic and always accompanied by validation metrics to avoid deterministic medical claims. By separating the preprocessing, segmentation, and prognostic layers into a modular REST-based backend, the system demonstrates a reproducible software lifecycle that supports future institutional retraining to prevent performance drift due to changing patient demographics or imaging hardware. The architecture aligns with IEEE Std 12207 [16] and ISO/IEC 25010 [17] for software lifecycle and quality, anticipates IEC 62304 [18] and FDA SaMD guidance [19] for medical-device software, and respects GDPR [20] and HIPAA [21] data protection principles.

E. Regulatory Alignment and Data Integrity

The development of NeuroAssist adheres to the IEEE Ethically Aligned Design (EAD) principles, specifically addressing transparency and algorithmic accountability. By providing the "AI Assistant" interpretability layer, the system avoids "black-box" predictions, instead offering clinicians a breakdown of tumor sub-region volumes in mm^3 . Furthermore, the data handling protocols were designed to be consistent with ISO/IEC 27001 standards for information security and the WHO guidance on Artificial Intelligence for Health, ensuring that patient data integrity is maintained throughout the ingestion and segmentation lifecycle.

VIII. SYSTEM ARCHITECTURE AND DEPLOYMENT

The NeuroAssist framework is implemented as a modular REST-based backend using the Flask web framework, enabling cross-platform inference capability. The deployment pipeline utilizes a decoupled architecture where the MRI preprocessing, U-Net inference, and survival regression modules communicate via structured JSON payloads. To ensure clinical utility, the system produces a "Three-View" visualization (Axial, Coronal, and Sagittal) alongside the 3D volumetric reconstruction. Performance benchmarks during deployment showed stable GPU utilization and deterministic inference times, confirming the system's readiness for integration into hospital Picture Archiving and Communication Systems (PACS) or specialized radiological workstations.

IX. CONCLUSION

This paper presented NeuroAssist, an integrated deep-learning framework for automated brain tumor analysis from multimodal MRI. The system combines a memory-efficient 2.5D U-Net segmentation network, a geometrically consistent volumetric reconstruction stage that produces tumor volumes in mm^3 , a regression-based survival estimator that couples volumetric descriptors with patient age, and a REST-based deployment backend that exposes all of the above through

a structured JSON contract. On the BraTS dataset, the segmentation network reached a peak validation Dice of 0.7559 and a mean IoU of 0.6457 at epoch 36, with class-wise Dice scores of 0.7617 (edema), 0.7743 (enhancing), and 0.6109 (necrotic core), indicating competitive spatial overlap under 2.5D constraints. End-to-end testing confirmed deterministic behavior, stable GPU inference, and coherent integration of segmentation, volumetric quantification, and survival estimation.

Future work will focus on migrating to full 3D convolutional architectures when larger GPU budgets are available, incorporating additional clinical, genetic, and longitudinal features to improve survival modeling, conducting external validation on independent multi-center hospital datasets, integrating explainable AI techniques to increase interpretability for clinicians, and optimizing the backend for scalable cloud-based deployment. The framework provides a reproducible and standards-aligned foundation on which more advanced clinical AI systems can be built.

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REFERENCES

- [1] B. H. Menze et al., "The multimodal brain tumor image segmentation benchmark (BRATS)," *IEEE Trans. Med. Imag.*, vol. 34, no. 10, pp. 1993–2024, 2015.
- [2] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.
- [4] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.
- [5] Ö. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, "3D U-Net: Learning dense volumetric segmentation from sparse annotation," in *MICCAI*, 2016, pp. 424–432.
- [6] K. Kamnitsas et al., "Efficient multi-scale 3D CNN with fully connected CRF for accurate brain lesion segmentation," *Med. Image Anal.*, vol. 36, pp. 61–78, 2017.
- [7] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. Int. Conf. 3D Vision (3DV)*, 2016, pp. 565–571.
- [8] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods*, vol. 18, pp. 203–211, 2021.
- [9] L. R. Dice, "Measures of the amount of ecologic association between species," *Ecology*, vol. 26, no. 3, pp. 297–302, 1945.
- [10] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal Visual Object Classes challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, 2010.
- [11] K. H. Zou et al., "Statistical validation of image segmentation quality based on a spatial overlap index," *Academic Radiology*, vol. 11, no. 2, pp. 178–189, 2004.
- [12] M. Brett et al., "NiBabel: Neuroimaging in Python," *J. Open Source Softw.*, vol. 5, no. 45, p. 1783, 2020.
- [13] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

- [14] L. Bottou, "Stochastic gradient descent tricks," in *Neural Networks: Tricks of the Trade*. Springer, 2012, pp. 421–436.
- [15] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, "Dissecting racial bias in an algorithm used to manage the health of populations," *Science*, vol. 366, no. 6464, pp. 447–453, 2019.
- [16] IEEE, IEEE Std 12207: Systems and Software Engineering - Software Life Cycle Processes, 2017.
- [17] ISO/IEC, ISO/IEC 25010: Systems and Software Quality Requirements and Evaluation, 2011.
- [18] International Electrotechnical Commission, IEC 62304: Medical Device Software Software Lifecycle Processes, 2015.
- [19] U.S. Food and Drug Administration, Software as a Medical Device (SaMD): Clinical Evaluation Guidance, 2017.
- [20] European Union, General Data Protection Regulation (GDPR), 2016.
- [21] U.S. Department of Health and Human Services, HIPAA Privacy Rule, 1996.